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Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 24-03-2026

Full-Stack Skill Learning Academy Marketplace with Secure Authentication

SDG: SDG No – SDG 4 – Quality Education

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- Full-Stack Skill Learning Academy Marketplace is an online learning platform.
- Provides a centralized system for students, instructors, and administrators.
- Enables secure registration and role-based login.
- Allows students to enroll in courses and track learning.
- Helps instructors create and manage courses.
- Reduces manual work and improves system efficiency.
- Built using full-stack technologies (HTML, CSS, Java, MySQL).
- Supports digital education under SDG 4 – Quality Education.
- Provides a user-friendly interface with modern design and smooth navigation.
- Ensures data security and efficient performance through proper backend integration.

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PROBLEM STATEMENT



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- Many educational systems rely on manual or outdated processes.

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- Lack of a centralized platform for managing courses and users.

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- Poor user experience due to disconnected systems.

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- Difficulty in tracking student progress and enrollments.

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- Limited automation in course management and administration.

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- Data redundancy and inefficiency in handling records.

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- Lack of secure authentication and role-based access.

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- Need for a scalable and efficient digital learning solution.

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OBJECTIVE(S)



- 1 ○ To develop a secure full-stack online learning platform with role-based access.
- 2 ○ To enable efficient course management, enrollment, and user tracking.

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PROPOSED SYSTEM



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- Provides a centralized web platform for students, instructors, and administrators.

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- Implements secure user authentication with role-based access.

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- Allows instructors to create and manage courses.

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- Enables students to enroll in courses and track progress.

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- Includes an admin dashboard for monitoring and analytics.

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- Uses a database system for efficient data storage and management.

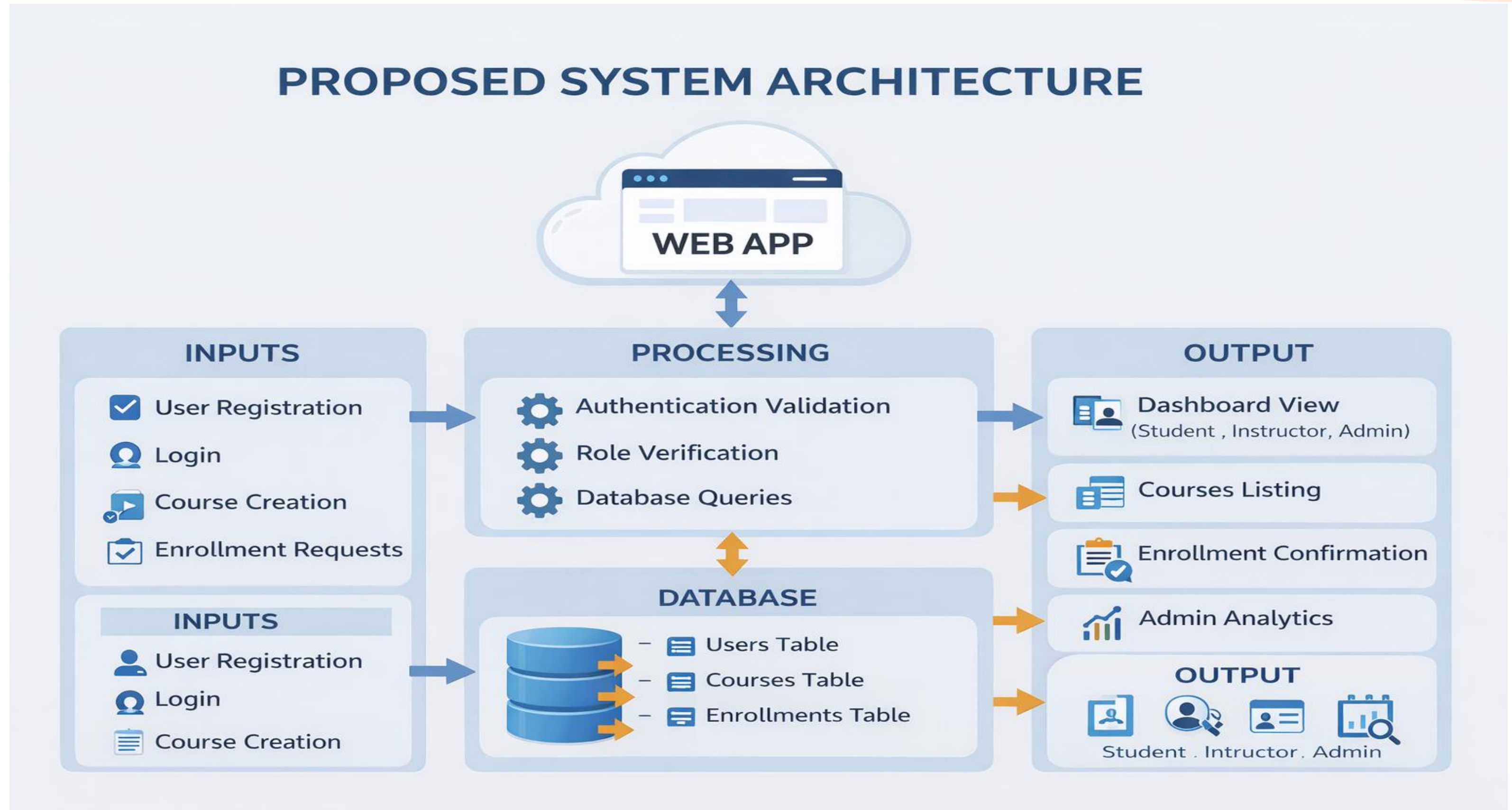
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- Prevents duplicate enrollments and ensures data consistency.

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- Provides a user-friendly interface with responsive design.

PROPOSED SYSTEM ARCHITECTURE



PROPOSED SYSTEM ARCHITECTURE

- Input →

User registration, login, course creation, enrollment requests.

- Processing →

Authentication validation, role verification, database queries.

- Database →

Users table, Courses table, Enrollments table, reviews table.

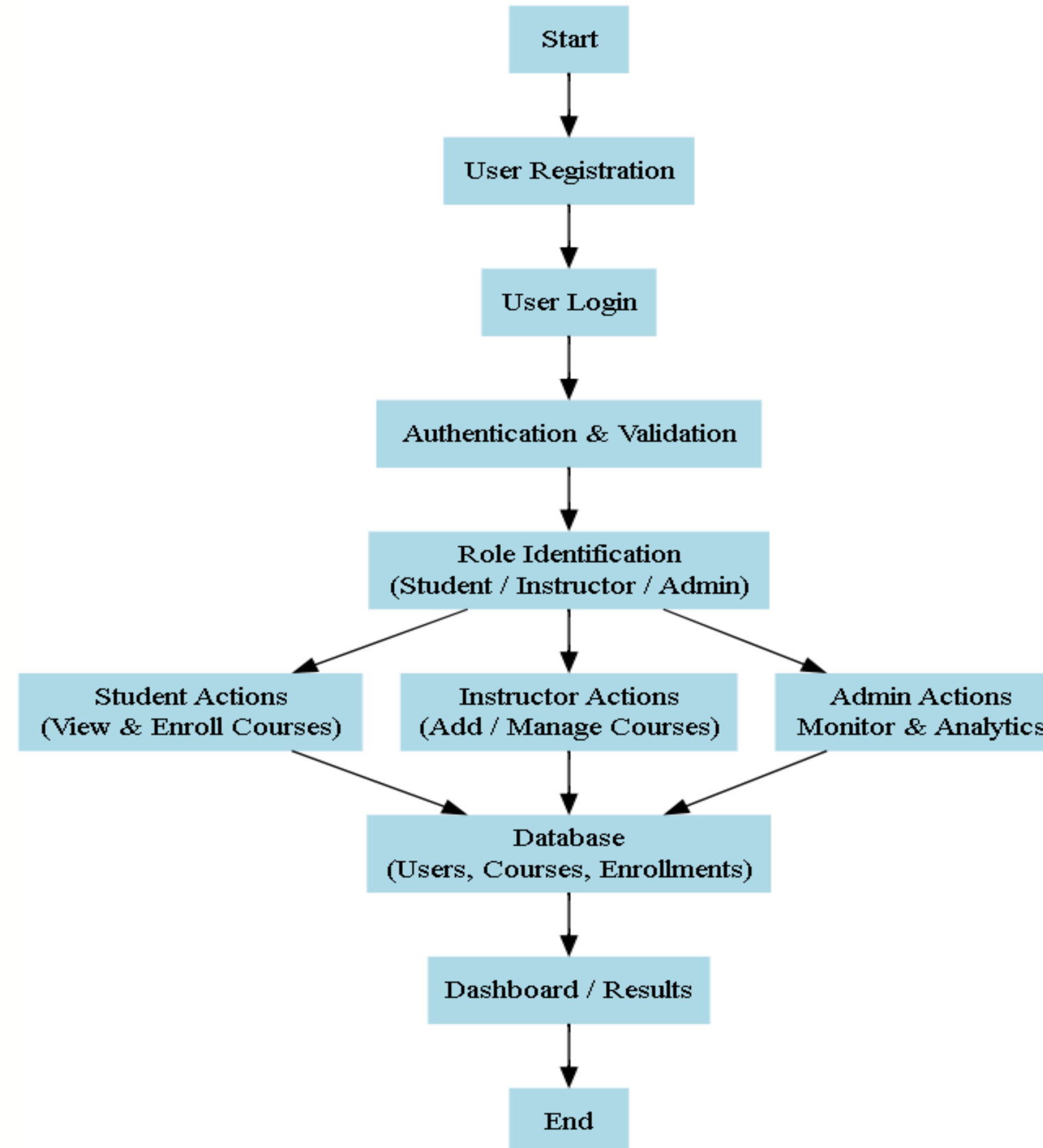
- Output →

Dashboard, course listings, enrollment confirmation, admin analytics.

WORKFLOW



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TECHNOLOGIES USED



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➤ **FRONTEND:**

- HTML
- CSS
- JavaScript

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➤ **BACKEND:**

- PHP

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➤ **DATABASE:**

- MySQL

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➤ **TOOLS:**

- XAMPP
- VS Code
- phpMyAdmin

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➤ **HARDWARE/SOFTWARE REQUIREMENTS:**

- System with minimum 4GB RAM
- Web browser (Chrome/Edge)
- XAMPP Server installed

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MODULE DESCRIPTION



➤ **Module 1: User Management**

- Handles user registration and login
- Implements secure password authentication
- Supports role-based access (Student, Instructor, Admin)
- Manages user profiles and details
- Ensures data validation and security

➤ **Module 2: Course Management**

- Allows instructors to add new courses
- Enables editing and deletion of courses
- Stores course details (title, description, content)
- Displays available courses to students
- Maintains course-related data in database

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MODULE DESCRIPTION



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➤ **Module 3: Enrollment System**

- Enables students to enroll in courses
- Prevents duplicate enrollments
- Tracks enrolled courses for each user
- Updates enrollment data in database
- Provides confirmation of enrollment

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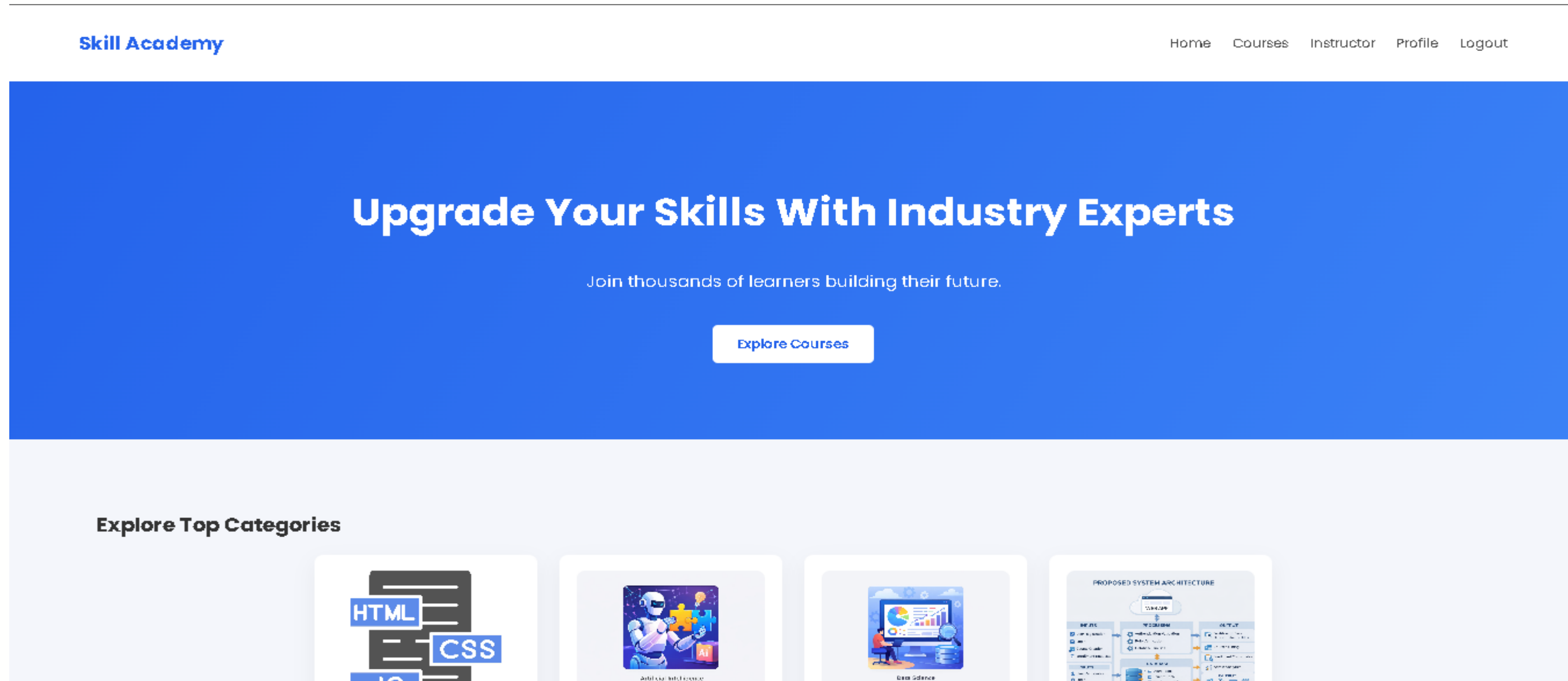
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➤ **Module 4: Admin & Analytics**

- Allows admin to view all users and courses
- Monitors enrollments and system activity
- Displays basic analytics and statistics
- Manages overall system operations
- Ensures smooth functioning of the platform

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IMPLEMENTATION SCREENSHOTS



➤ 🏠 Home Page

- Displays the main landing page with platform introduction and featured courses.
- Provides navigation to courses, instructor section, profile, and logout.

IMPLEMENTATION SCREENSHOTS



Skill Academy Home Login Register

Login

Skill Academy
Upgrade your skills with industry experts.

Platform
Courses
Instructors
Certificates

Security
SSL Protected
Secure Login
Encrypted Data

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Designed by **M INDRAJITH**

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➤ Login Page:

- Provides a secure login interface where users enter email and password to access the system.
- Validates user credentials and redirects based on role (Student/Instructor/Admin).

IMPLEMENTATION SCREENSHOTS



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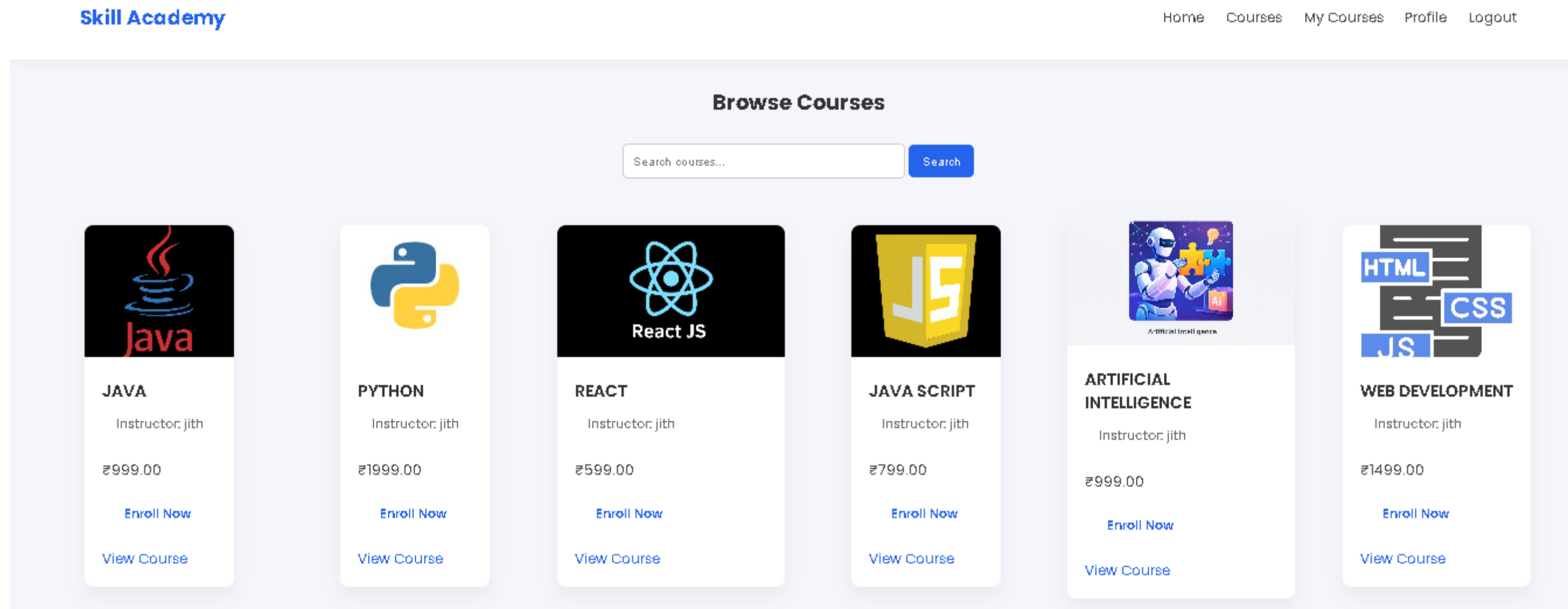
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➤ 📖 Course Listing Page

- Displays all available courses with details like title, instructor, and price.
- Allows users to browse and select courses easily.

IMPLEMENTATION SCREENSHOTS



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Course Payment

JAVA

Price: ₹999.00

Card Holder Name

Card Number

Expiry Date

MM/YY

CVV

[Pay Now](#)

Payment Successful

Your course has been successfully enrolled.

[Go to My Courses](#)

➤ **Course Payment Page**

- Displays course details and allows users to enter payment information like card details.
- Processes user payment securely before confirming enrollment.

➤ **Payment Success Page**

- Confirms that the payment is completed and the course is successfully enrolled.
- Provides option to navigate to “My Courses” section.

ADVANTAGES & LIMITATIONS



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➤ ✓ **Advantages:**

- Provides a centralized online learning platform.
- Secure authentication with role-based access.
- Easy course management for instructors.
- Simple enrollment process for students.
- Improves efficiency and reduces manual work.

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➤ ✗ **Limitations:**

- Requires internet connection to access.
- Limited advanced features (like AI recommendations) .
- Basic payment system (can be improved).
- UI can be enhanced further.
- Not available as a mobile application.

Thank you

